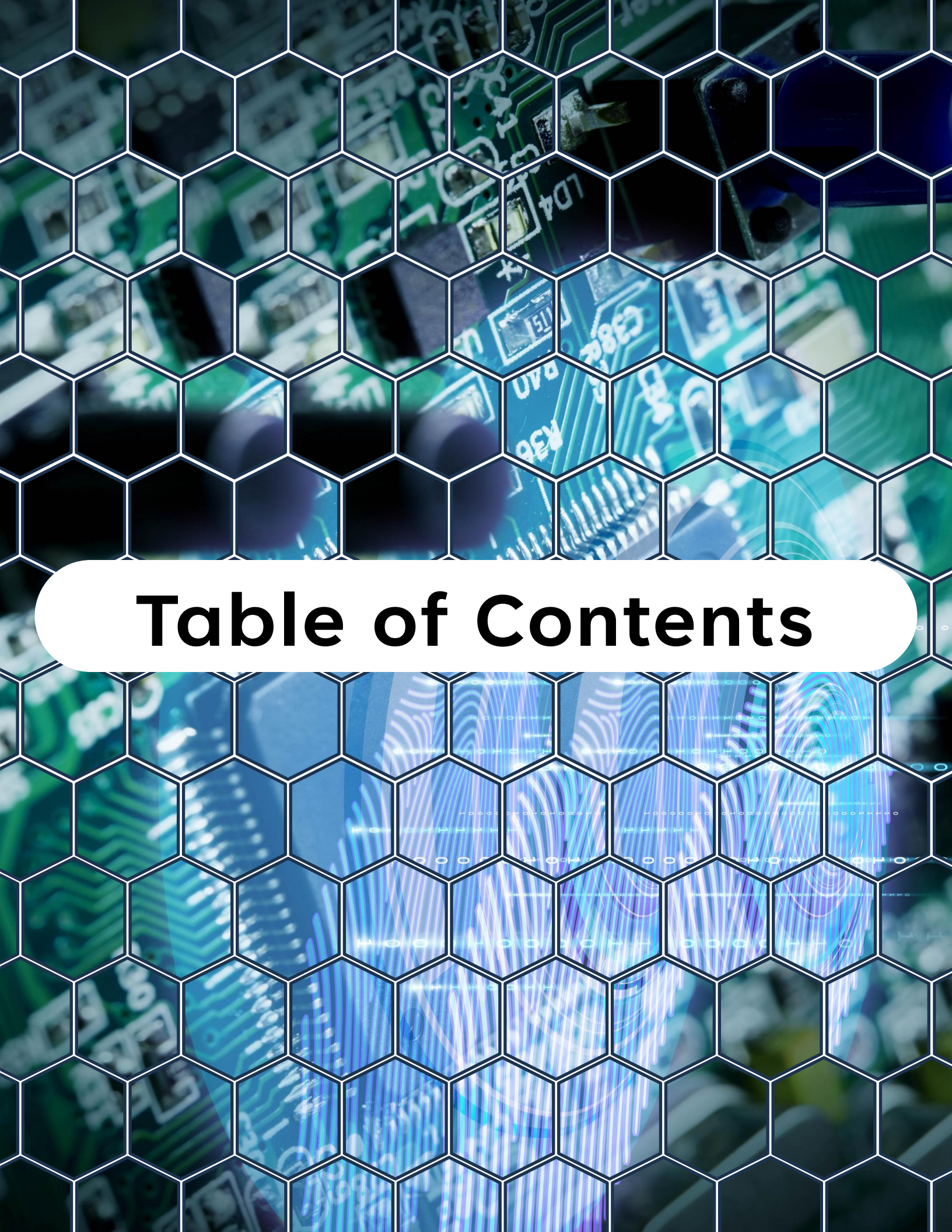
30-Day FreeRTOS Course for ESP32 Using ESP-IDF

(Day 19)

“Memory Management in FreeRTOS”



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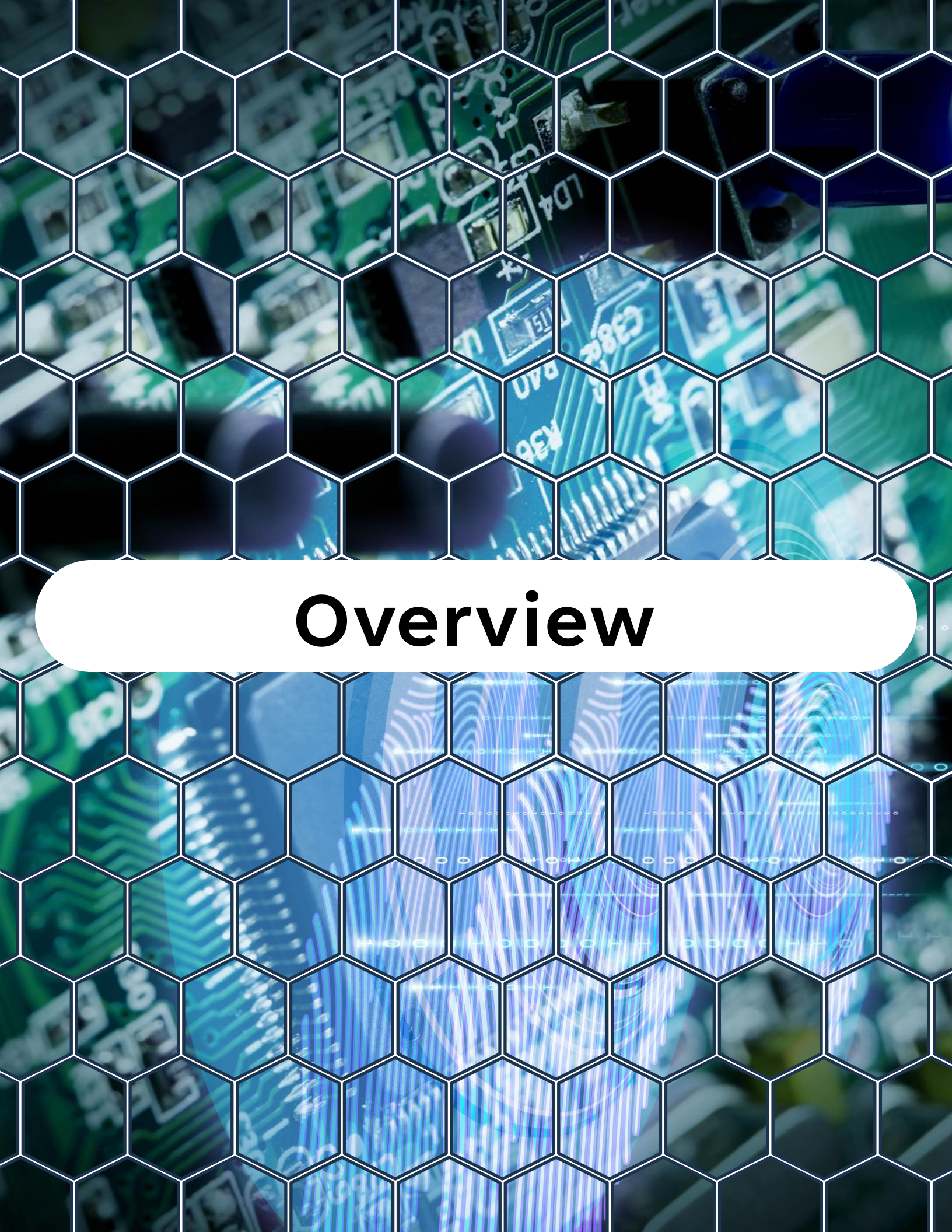


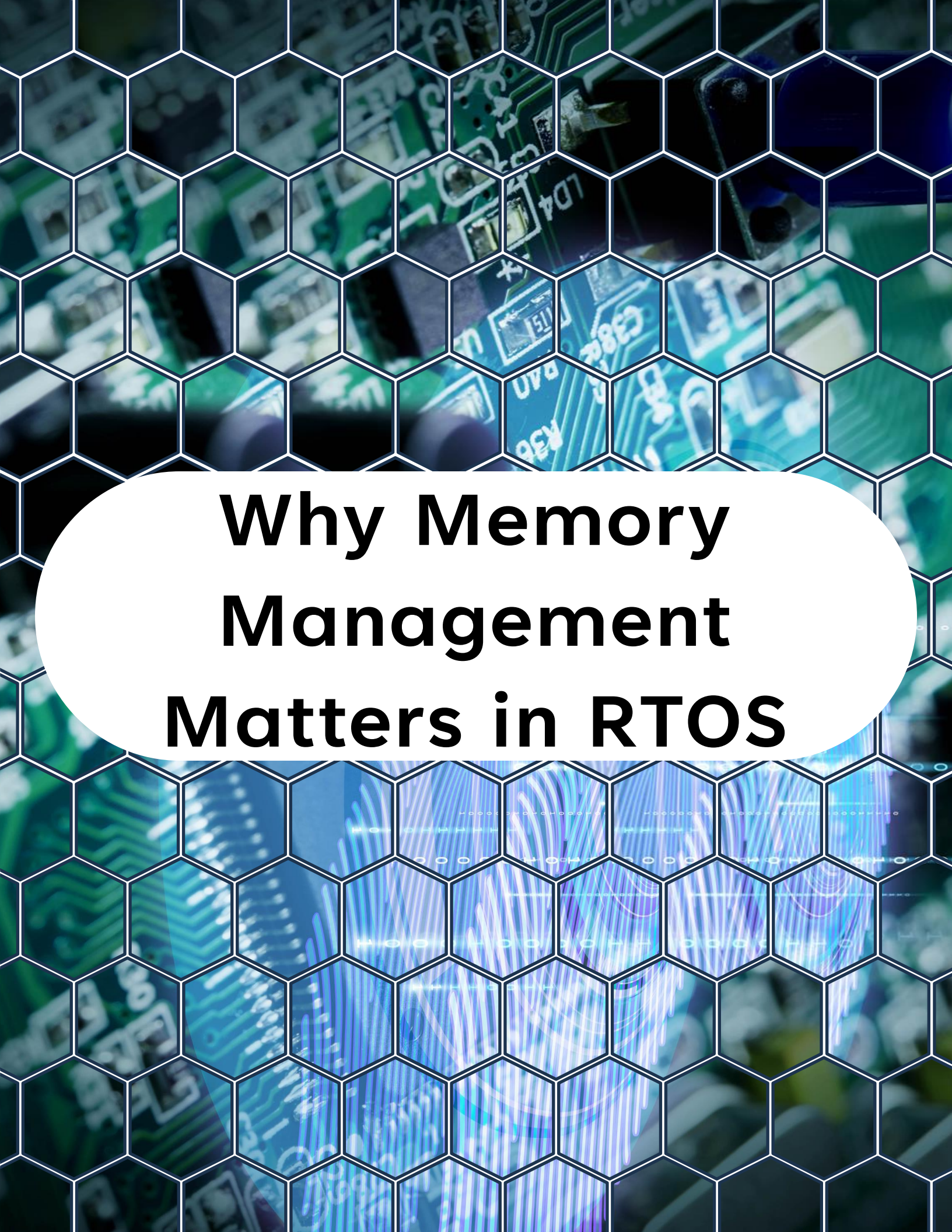
Overview

Overview

On **Day 19**, you'll learn:

- How FreeRTOS manages memory on embedded systems
- The role of the **heap** in task stacks, queues, semaphores, and timers
- The different **heap memory allocation schemes (heap_1 to heap_5)**
- How to configure FreeRTOS memory management in ESP-IDF
- How to use **static allocation** instead of dynamic allocation
- Best practices for efficient and safe memory use on ESP32

By the end of this lesson, you'll understand how to **allocate, manage, and optimize memory** in FreeRTOS-based ESP32 applications.



Why Memory Management Matters in RTOS

Why Memory Management Matters in RTOS

Unlike PCs, embedded systems like ESP32 have **limited RAM** (e.g., 320 KB internal RAM + optional external PSRAM).

Poor memory management can lead to:

- **Heap fragmentation**
- **Task creation failures**
- **Unpredictable system crashes**

📌 FreeRTOS provides flexible heap allocation schemes so you can **balance performance and safety for your application**.

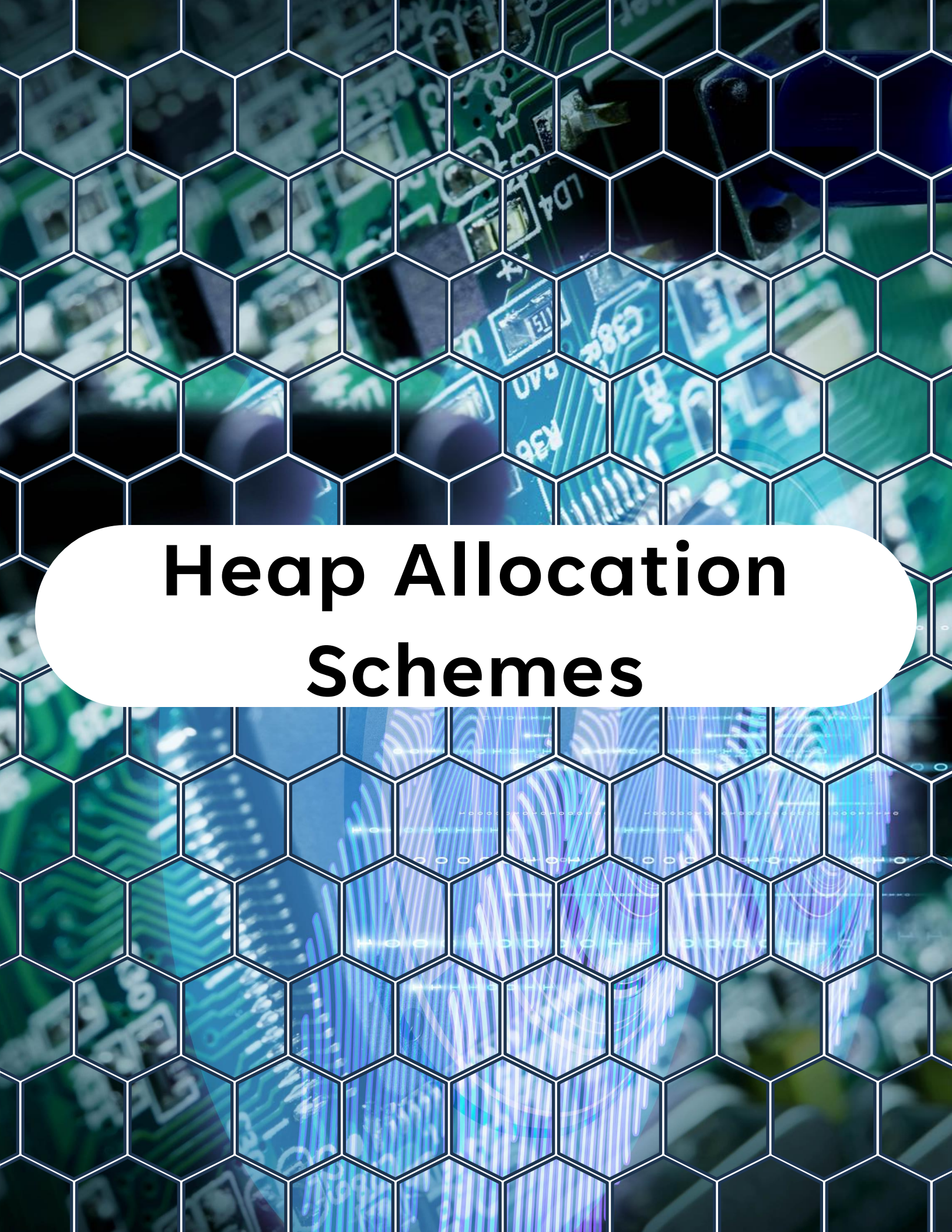
The Heap in FreeRTOS

The Heap in FreeRTOS

The heap is a global memory pool used for:

- Creating tasks (**xTaskCreate**)
- Creating queues, semaphores, and event groups
- Creating software timers

FreeRTOS does not use the standard `malloc()/free()` from libc. Instead, it provides its own heap allocators, selected at compile time.



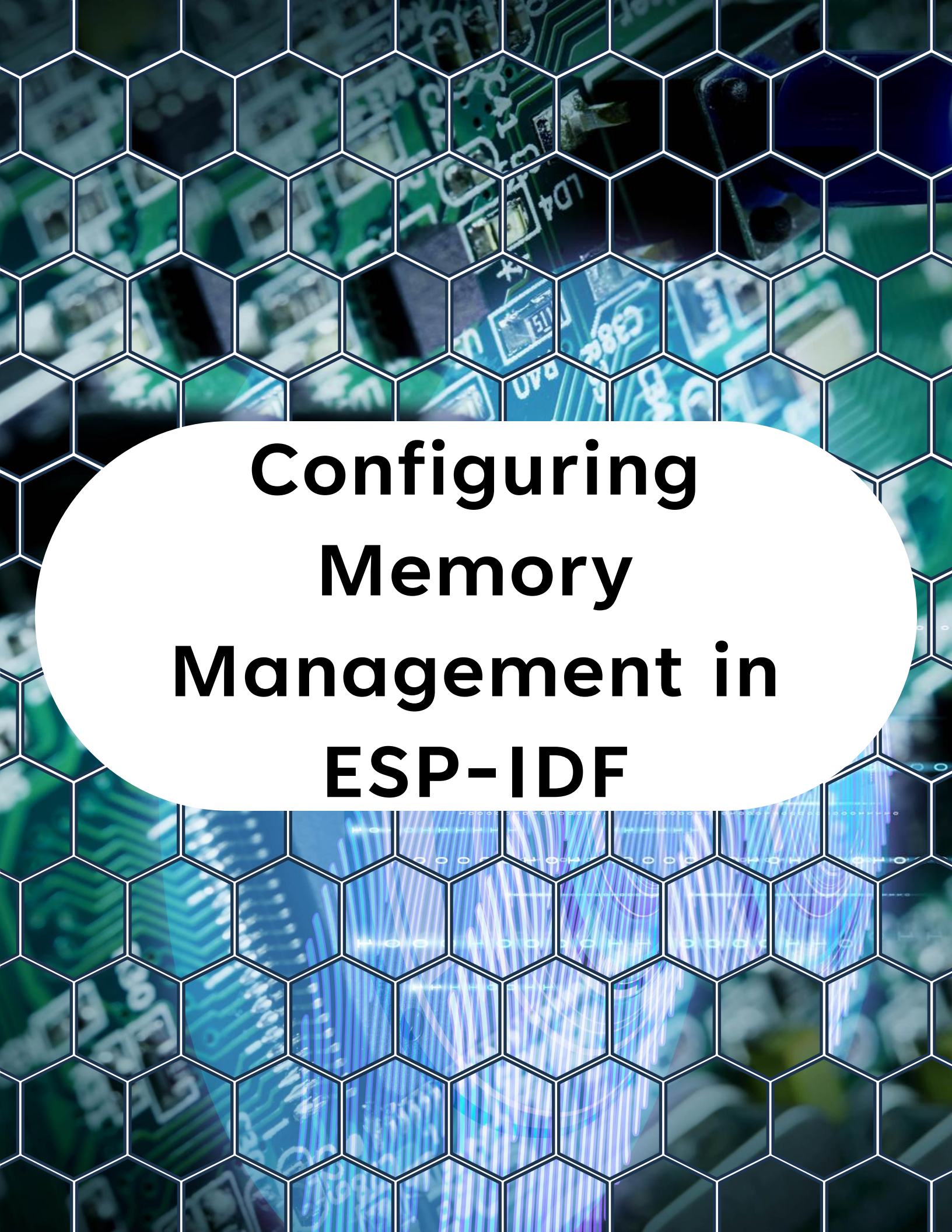
Heap Allocation Schemes

Heap Allocation Schemes

FreeRTOS offers five heap implementations in `heap_x.c`:

Scheme	Description
heap_1	Simplest, only malloc(). No free(). No fragmentation.
heap_2	Allows free(). Simple algorithms but can fragment.
heap_3	Uses standard C malloc()/free(). Depends on system allocator.
heap_4	Best balance. Allows free(), uses coalescing to reduce fragmentation.
heap_5	Like heap_4, but supports multiple memory regions (e.g., internal + external RAM).

📌 ESP-IDF typically uses heap_4 by default, as it handles fragmentation well.


Configuring
Memory
Management in
ESP-IDF

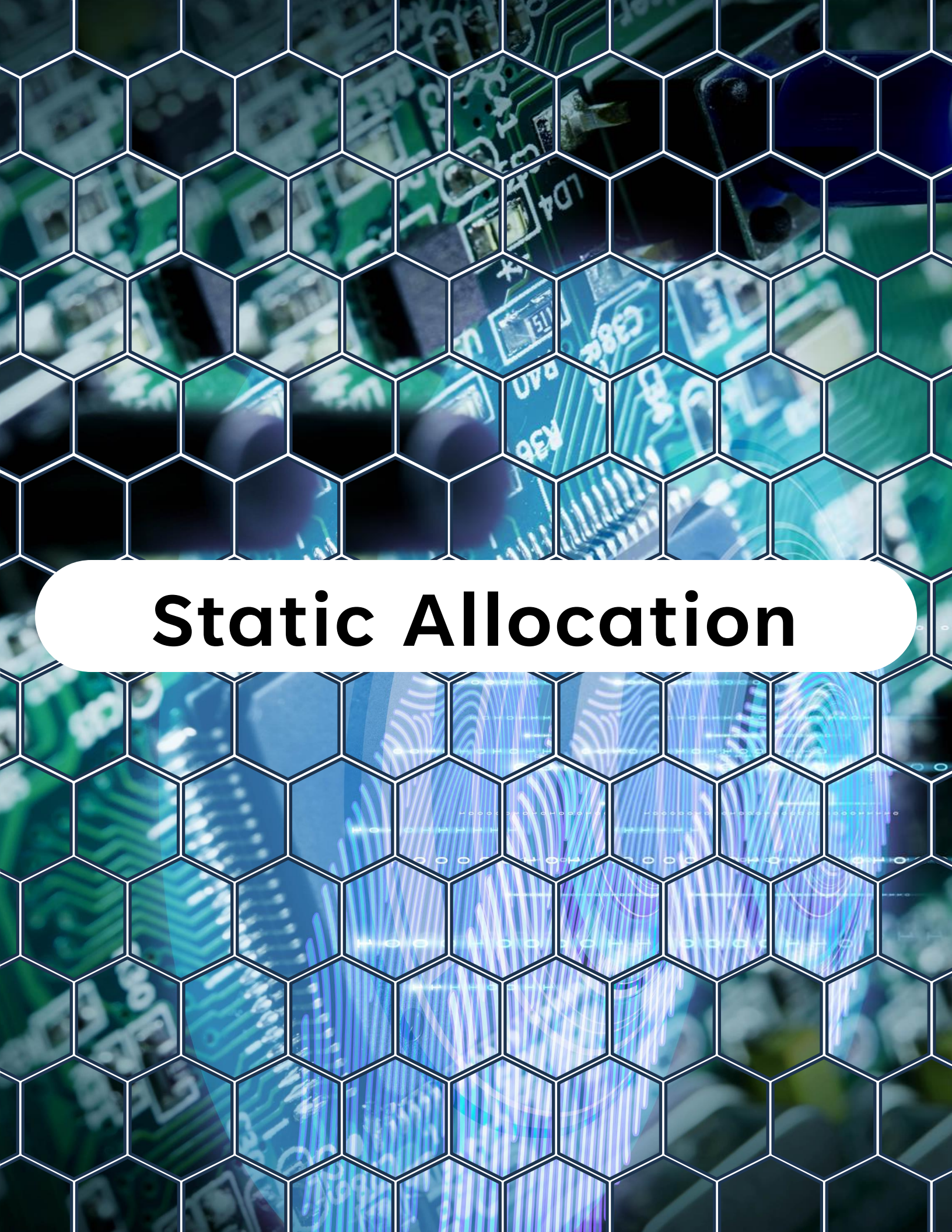
Configuring Memory Management in ESP-IDF

In ESP-IDF, memory allocation settings are controlled via:

`idf.py menuconfig → Component config → FreeRTOS → Heap memory`

You can:

- Select which `heap_x.c` implementation to use.
- Enable **static allocation support**.
- Configure **stack overflow checking**.
- Enable **heap poisoning** for debugging memory corruption.
- Force the entire heap component to be **placed in flash memory**



Static Allocation

Static Allocation

Instead of using dynamic allocation (`xTaskCreate`), FreeRTOS also supports static allocation (`xTaskCreateStatic`).

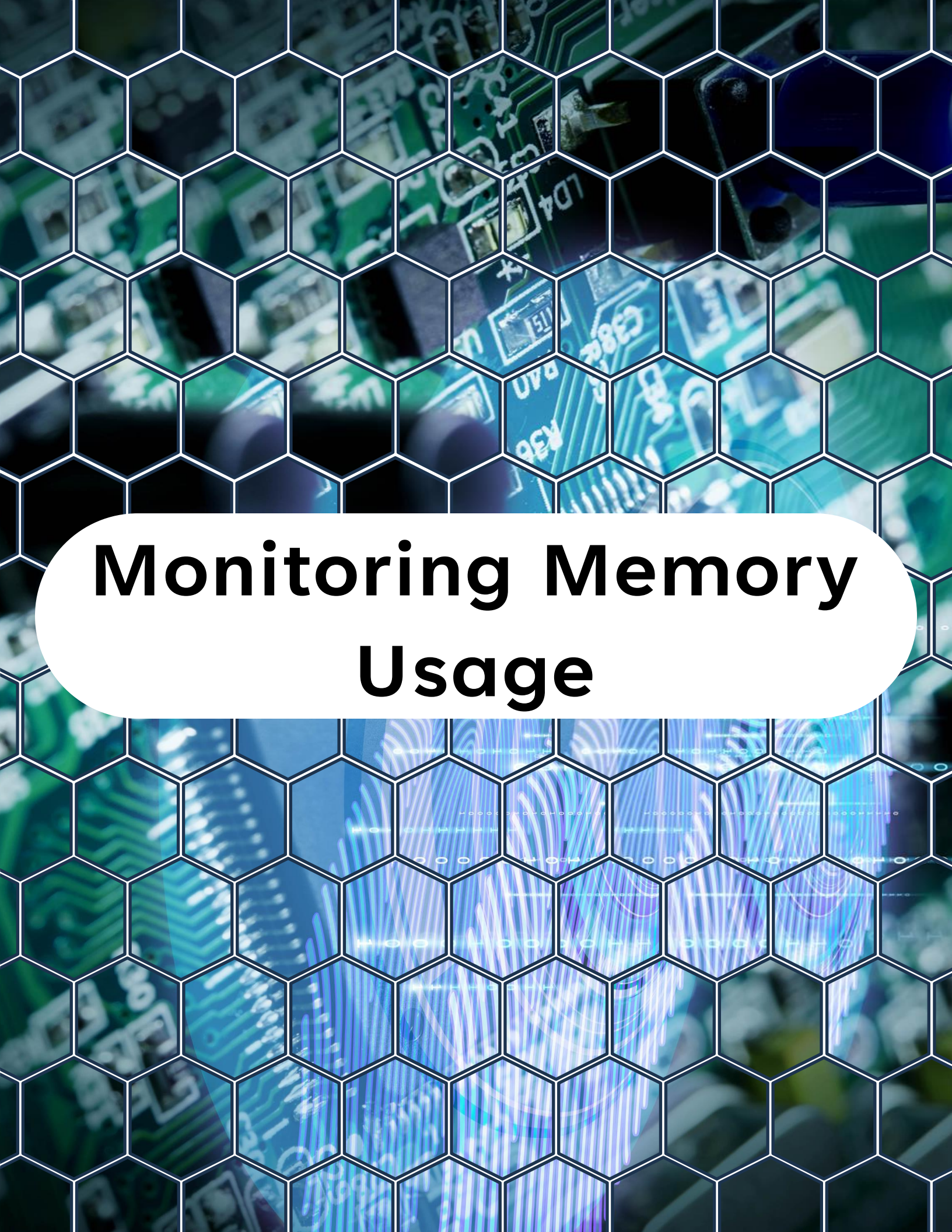
Example – Statically Allocated Task

```c
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "esp_log.h"

#define TAG "static-demo"

// Start around 2-3 KB for simple tasks that call library
// code or printf(), then tune using
// uxTaskGetStackHighWaterMark2()
#define STACK_SIZE_BYTES (3 * 1024)

// Task control block (TCB) for the task
StaticTask_t task_buffer;

// Stack memory for the task
StackType_t task_stack[STACK_SIZE_BYTES];

// Task function
void static_task(void *pvParameters) {
    while (1) {
        ESP_LOGI(TAG, "Running static task; free stack (bytes): %u",
                 (unsigned)uxTaskGetStackHighWaterMark2(NULL));
        vTaskDelay(pdMS_TO_TICKS(1000));
    }
}

// Main application entry point
void app_main() {
```

Static Allocation

```
27 // Main application entry point
28 void app_main() {
29     // Create task without using the heap
30     TaskHandle_t handle = xTaskCreateStatic(
31         static_task,      // Task function
32         "StaticTask",     // Task name
33         STACK_SIZE_BYTES,       // Stack size
34         NULL,             // Parameters
35         5,                // Priority
36         task_stack,       // Stack memory
37         &task_buffer      // Task control block
38     );
39 
40     if (handle == NULL) {
41         ESP_LOGE(TAG,
42             "xTaskCreateStatic failed (insufficient stack/invalid args)");
43     }
44 }
```

No heap allocation → predictable, safer for memory-constrained applications.



Monitoring Memory Usage

Monitoring Memory Usage

FreeRTOS provides APIs to check memory usage:

- `xPortGetFreeHeapSize()` → total free heap.
- `xPortGetMinimumEverFreeHeapSize()` → lowest heap ever recorded (good for tuning).

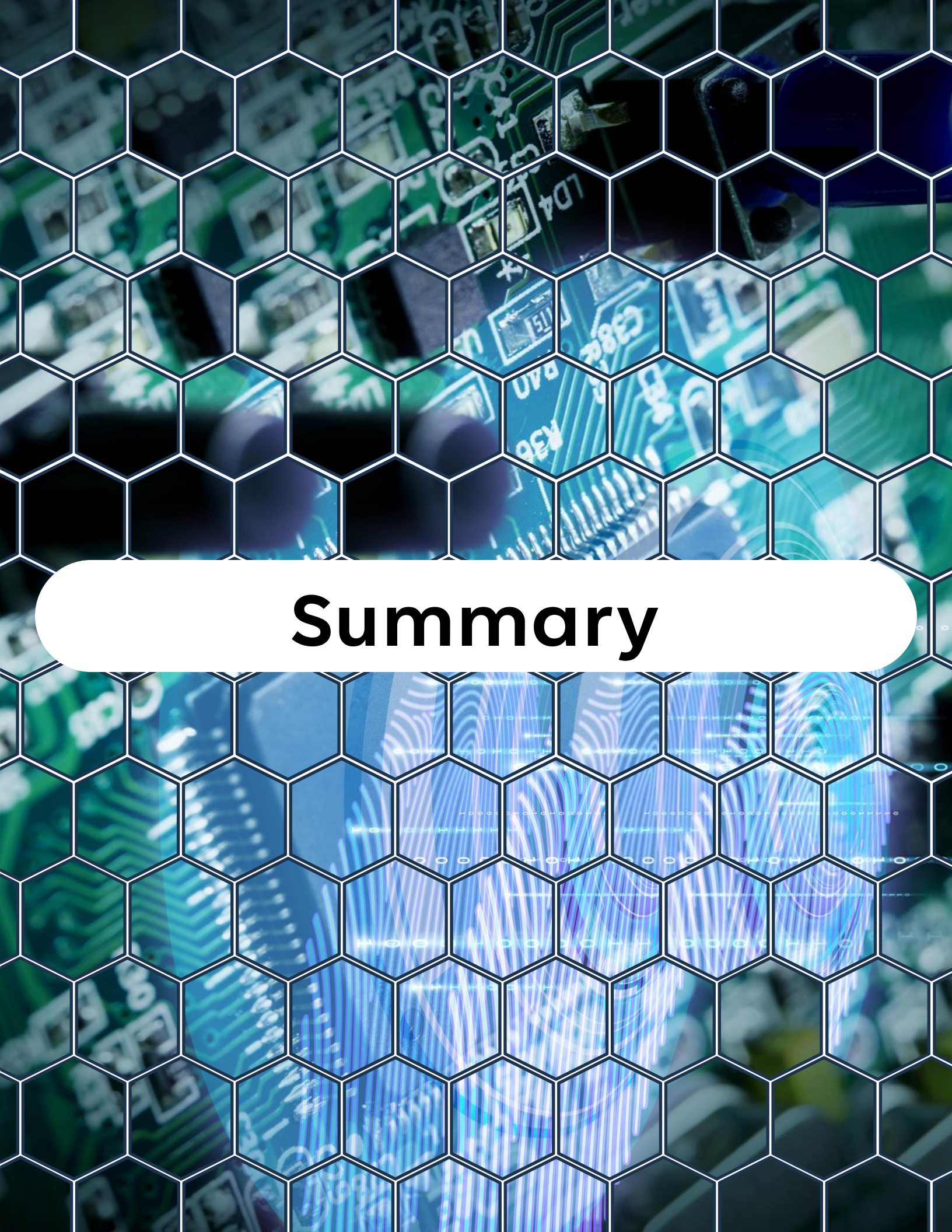
ESP-IDF also includes:

- `heap_caps_get_free_size(MALLOC_CAP_8BIT)` → free memory of a given capability.
- `heap_caps_print_heap_info(MALLOC_CAP_8BIT)` → detailed heap usage.

Best Practices

Best Practices

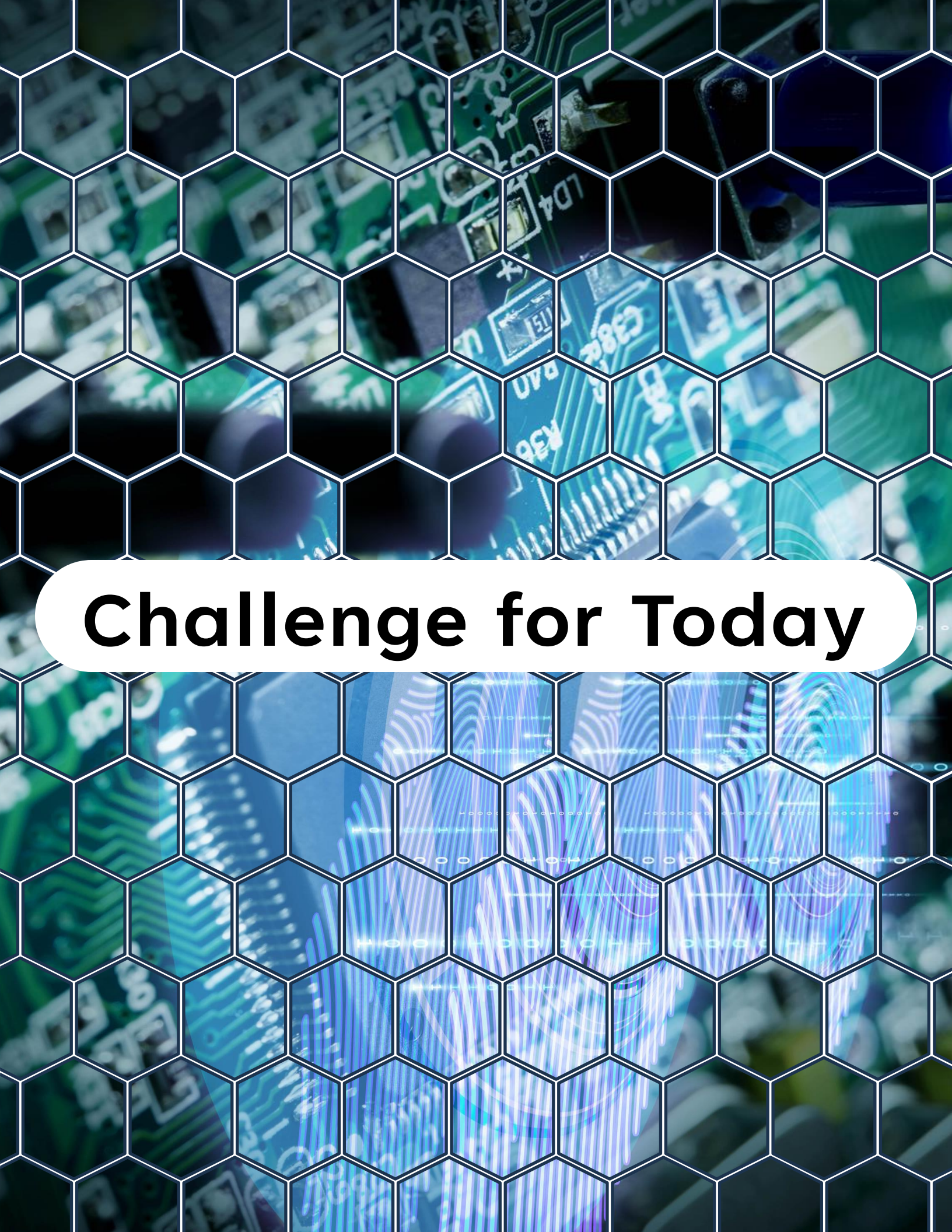
- ✅ Use **heap_4** (or **heap_5** if using multiple regions).
- ✅ Prefer **static allocation** for critical tasks (predictable).
- ✅ Regularly check `xPortGetMinimumEverFreeHeapSize()` to ensure enough headroom.
- ✅ Keep task stacks small and measured — use `uxTaskGetStackHighWaterMark()` to tune.
- ✅ For heavy data, use `external PSRAM` (on ESP32-WROVER or S3 with PSRAM).
- ✅ Enable **stack overflow detection** in menuconfig.



Summary

Summary

- FreeRTOS memory is managed through **heap_x.c** implementations.
- ESP-IDF defaults to **heap_4**, which is safe and efficient.
- Use **static allocation** when predictability is required.
- Monitor heap and stack usage to prevent crashes.
- Use PSRAM or heap_5 if your app requires multiple memory regions.


Challenge for Today

Challenge for Today

🛠️ Create two tasks:

- One using xTaskCreate (dynamic).
- One using xTaskCreateStatic (static).
- Print memory statistics using xPortGetFreeHeapSize() before and after creation.
- Compare the difference and discuss which approach is safer for real-time systems.

"Thanks for watching!

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